

**PATLOLLA SAIRAM REDDY**

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## PROFESSIONAL SUMMARY

GCP Data Engineer with 4.5 years of overall IT experience, including hands-on experience in designing scalable data pipelines and working with AI/ML-based data processing workflows. Skilled in Python, SQL, and PySpark with experience in NLP, prompt engineering, LLM workflows, and data processing. Experienced in BigQuery, Dataflow, Cloud Storage, and Airflow (Cloud Composer), with strong focus on performance optimization, data quality, and scalable system design. Also brings experience in building APIs and integrating data platforms with AI-driven solutions.

## WORK EXPERIENCE

**PRODAPT SOLUTIONS Private Limited** –  November 2025 – Present

**INFOSYS LIMITED** -  September 2021 –  October 2025

### SENIOR SOFTWARE ENGINEER

- Designed and implemented scalable ETL pipelines for structured and unstructured data using Python, BigQuery, Dataflow, and Cloud Composer.
- Developed end-to-end data ingestion and transformation workflows supporting AI applications, and enterprise analytics use cases.
- Built Python-based data processing modules for handling large-scale text data supporting NLP and LLM based applications.
- Created and optimized complex SQL queries and pipelines, improving performance and reducing processing cost.
- Worked with structured and unstructured datasets for AI/ML applications including classification, retrieval and text processing.

## SKILLS

|                               |                                                                                       |
|-------------------------------|---------------------------------------------------------------------------------------|
| <b>Cloud Platforms</b>        | Google Cloud Platform (GCP)                                                           |
| <b>GCP Services</b>           | BigQuery, Google Cloud Storage, Cloud Composer (Airflow), Pub/Sub, Dataproc, IAM      |
| <b>Languages</b>              | SQL, Python, PySpark, React, FastAPI                                                  |
| <b>AI / ML &amp; LLM</b>      | NLP, Embeddings, Vector Retrieval, Model evaluation, Data processing, AI/ML workflows |
| <b>Architecture</b>           | RAG Architecture, Agentic workflow, Solution design, Microservices                    |
| <b>Workflow Orchestration</b> | Apache Airflow, Cloud Composer                                                        |
| <b>LLM Frameworks</b>         | LangChain, LangGraph, LangSmith, RAG, OpenAI SDK, CrewAI                              |
| <b>Project Tools</b>          | Agile(Scrum, Sprint), Jira, Confluence, GitHub                                        |

## PROJECTS

**Project Name:** AIOps

**Client Name:** Verizon Communications – Telecom & Digital Services

**Project Description:**

Verizon is a leading global telecommunications company providing wireless services, broadband, and digital solutions. This project involved migrating large-scale sales, customer, and network usage data from legacy systems to Google Cloud Platform. The solution enabled near real-time data availability in BigQuery, supporting downstream machine learning models for predictive, forecasting, and statistical analytics and providing curated datasets for API-driven applications.

**Roles and Responsibilities:**

- Worked on LLM workflows including prompt engineering, dataset preparation (JSONL), evaluation techniques like RAGAS and DeepEval, AI-ready data processing.
- Developed Python-based APIs using FastAPI for enterprise data services and AI applications.
- Supported near real-time data pipelines enabling AI-driven insights and predictive analytics.
- Designed AI-ready datasets through data cleansing, transformation, and feature preparation for ML and LLM workflows.
- Contributed to building data pipelines supporting AI/ML models for anomaly detection and predictive analytics use cases.
- Built retrieval-based data pipelines leveraging embeddings and vector representations for intelligent search applications.
- Optimized BigQuery cost and performance by tuning SQL/stored procedures.

**Project Name:** Enterprise Data Warehouse Modernization & Advanced Analytics [EDWM]

**Client Name:** Charles Schwab (Leading U.S. Financial Services Firm)

**Project Name:**

Charles Schwab is one of the largest investment and brokerage firms in the United States, providing a wide range of financial services including wealth management, trading, and retirement planning. Migrated Schwab's legacy on-premises enterprise data warehouse to Google BigQuery to support scalable, secure, and real-time financial analytics. Designed and implemented data pipelines to ingest trading, customer, and transaction data from multiple systems into GCP.

**Roles and Responsibilities:**

- Processed large-scale unstructured and semi-structured data to support AI/ML and analytics use cases.
- Developed SQL and Python data transformation scripts for cleansing, deduplication, and aggregation.
- Worked with transformer-based models and explored prompt engineering, embeddings, and LLM integration using open-source frameworks.
- Automated analytics workflows using Cloud Composer for daily, weekly, and monthly processes.
- Implemented retrieval-based data pipelines (RAG concepts) for enabling context-aware AI-driven insights.
- Designed and optimized partitioned & clustered BigQuery tables for high-performance querying and created business-ready views in BigQuery for analytics teams.

## EDUCATION

 **B.Tech. in Computer Science & Engineering with 8.67CGPA**

 B V Raju Institute of Technology, Narsapur